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24EC2210 Network protocol & security HarshaVardhan

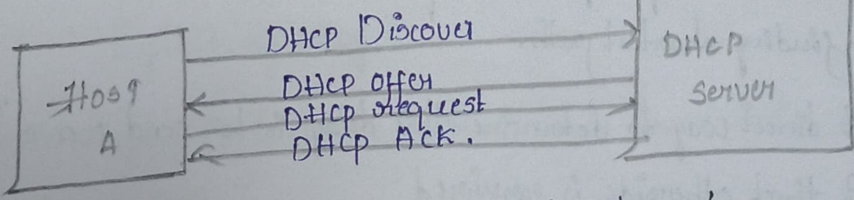
1) Illustrate the operation of DHCP with a neat Diagram?

Dynamic host Configuration protocol (DHCP) is a network protocol that automatically gives device an Ip adderess and other network settings so they can Connect and Communicate on a network without manual Configuration.

- Assigns unique Ip address to each device.
- Provide the Subnet mask to define the local network range.
- Configure default gateway for Communication outside local network.

DHCP operation :→

1) DHCP Discover : The client broadcast a request to find the DHCP
2) DHCP offer : The DHCP Server respond with an available Ip address and Configuration details (subnet, mask gateway)
3) DHCP request : The client requests the offered Ip address from the Server
4) DHCP Ack : The Server Confirms and allocates the Ip address to the client.



2) Analyze the Tcp Connection esthablishment state diagram and identify the key Concept involved in transferring data!

The tcp Connection establishment state diagram illustrate the process of establishment a Tcp Connection between a client and a server. key Concept involved in transferring data include in three way handshaking which ensure reliable data transfer and the Various states that the Connection goes through such as SYN, SYN-Ack, Ack these State represent different stages of the Connection, from initial request to ack.

and actual transfer of data. the diagram also highlight the importance of option negation during the handshake process. which Can include parameter like window size.

3) Construct steps involved in link state routing algorithm?

- Each router must be follow this five steps:
  1, Discover its neighbours, learn their network address.
  2, Measure the delay or cost to each of its neighbour.
  3, Construct a packet telling all it just learned.
  4, Send this packet to all other router.
  5, Compute the shortest path to every other routers.

1, Learning about neighbour:

- when a router is booted, its first task is to learn.
- it accomplishes this goal by sending a special hello packet
- The router on the other end is expected to send back a reply
- These names must be globally unique.

2, Measuring linecost:

- The link state routing algorithm requires each link to have a distance or delay for finding shortest path.
- The most direct way to determine this delay is to send over a line a special ECHO packet that otherside is required

3, Building linkstate Packets

- The identity of the Sender.
- Sequence number
- Age.
- List of neighbour and cost to each neighbour
- Building link state packets easy.

4) classify different Congenstion Control techniques and analyze leaky bucket and Token bucket traffic shaping technique!

open loop Congenstion Control :→

Aim: Prevent Congenstion before it occur.

No feedback mechanism from the network.

Techniques involved :→

- Retransmission policy
- Window policy
- Acknowldgement policy
- Discarding policy
- Admission Control.

closed loop Congenstion Control :→

Aim: Detect and react to Congenstion.

uses feedback from the network.

Techniques included:

- Back pressure.
- choke packets
- implicit signaling
- Explicit signaling.

Traffic shaping Control the rate of data transmission to avoid Congenstion

Two key techniques

1) leaky bucket Algorithm

2) Token bucket Algorithm

1) leaky bucket algorithm:

- Imagine a bucket with a small hole at the bottom.
- Incoming packet are stored in bucket (queue).
- Packet are transmitted at a Constant rate.
- If the bucket overflow.

2) Token bucket algorithm:

- Token are generated at a fixed rate and stored in bucket.
- To send a packet, the Sender must Consume tokens.
- If enough token are available packet is sent immediately.
- If not packet waits.

5) Apply the Concept of zones and domain DNS by demonstrating their use in resolving a Sample domain name?

The "Domain Name System" is a hierarichal System used to translate human-redable domain name into ip address. in DNS, the Concepts of domain and zones are essential for Organizing and managing this hierarchy. A domain represent a logical structure of subtree within the DNS namespace, such as [example.com], which may further include subdomain like [www.example.com]. on the other hand a, zone in adminstrative. portion of the domain space that is managed by specific DNS Server and Contain resource records. while a domain define the name hierarcy, a zone define how that hierarchy is distributed and Controlled accross different Server.

6) Discover the function of the user Datagram protocol in the transport layer!

The user Datagram protocol is a simple Connectionless transport layer protocol in the internet protocol suite, defined RFC 768 its primary function is to provide fast and efficient data transmission without establishing a Connection between sender and reciever Unlike Tcp, UDP does not guarantee delivery ordering or error Correction, making it lightweight and suitable for time-Sensitive application it operates by sending independent packet is called Datagrams, allowing quick Communication with minimal overhead.

1) Connectionless Communication.

2) Fast Datatransmission.

3) Multiplexing and demultiplexing.

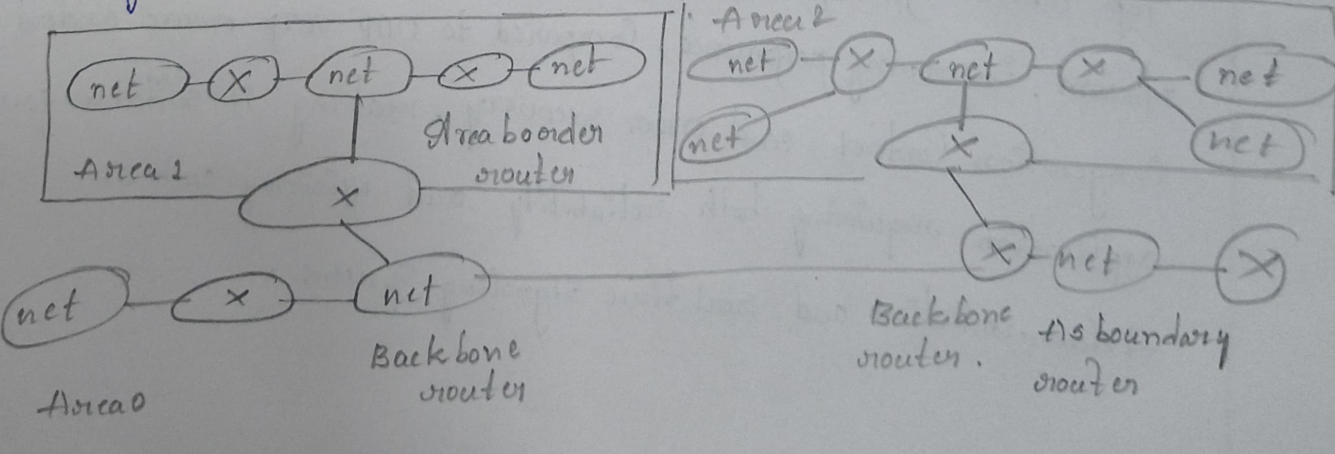
7) Illustrate the function of Routing information protocol in network layer?

Ans: Routing information protocol is a distance Vector routing protocol that operates at the network layer to determine the best path for data transmission between network. it uses the Bellaman Ford algorithm to calculate shortest path based on hop Count, where each router between Source and destination is Considered one hop. RIP router periodically exchange their routing table (every 30 seconds) with neighbouring router to maintain update network information. this protocol select the route with the minimum number of hop, maximum limit of 15 hops any route beyond this is Considered unreacheable. RIP also include mechanism like split horizon.

8) Show OSPF operates at network layer by working through a route selection ex;

Open shoortest path first is a link state routing protocol that operates at the network layer determine the most efficient path for data transmission within an autonomous System. unlike distance Vector protocol, OSPF builds a Complete map of network by exchanging link state advertisement (LSA) among router Each router that applies "Dijkstras Algorithm" to Compute the shortest path tree Selecting route based on cost (which depends on bandwidth). this allow OSPF to quickly.

Diagram :→



9) Abbreviate Simple mail transfer protocol (SMTP) in internetworking and explain it significance in ensuring reliable email Communication accross network?

Simple mail transfer protocol is an application layer protocol used in internetworking to send and relay emails across network. it define how mail server Communicate with each other to transfer message from the Sender's mail Server to recipient's mail server the internet. SMTP work on a store and forward mechanishm, ensuring that emails as tempararily stored and retransmitted until they reach the destination Successfully. it uses Tcp (typically port 25, 465 or 587) to provide reliable delivery. SMTP also incorporates error handling acknowldgement response, and retry mechanism, which help ensure that email message are delivered accurately and efficiently accross different network.

10) Analyze the functions of stream Control transmission protocol in the transport layer by Comparing it with Tcp and UDp?

stream Control transmission protol (SCTP) is a transport layer protocol that Combine reliability of transmission Control protocol with Some efficiency features of user datagram protocol it provide reliable, message-oriented Communication with advanced Capabilities such as multistreaming and multihoming unlike Tcp which Suffer from head-of-line blocking SCTP allow parllel delivery of stream, improving perfomance, Compared to UDP SCTP ensure orderd delivery Congenstion Control and error recovery, making it suitable four application requiring both reliability and efficiency such as telecomunication and real time signaling Systems.